Benjamin Brown

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Education

Harvey Mudd College, Claremont, CA, *B.S., Engineering, Emphasis in Environmental Analysis*
Expected May 2028

- GPA: 3.582, Dean's List
- Relevant coursework: Experimental Engineering, Sustainability and Resiliency in the Built Environment, Applied Math for Engineering, Engineering Systems (with practicum), Intro to Engineering Design and Manufacturing (with lab), Chemical and Thermal Processes, Probability and Statistics for Engineers, Linear Algebra, Mechanics and Wave Motion, Intro to Computer Science (with lab), Single & Multivariable Calculus

Punahou School, Honolulu, HI, *High School Diploma*
June 2024

- GPA: 3.856, Honors
- President's Award Recipient

Skills

Software & Programming: SolidWorks (CAD), AutoCAD, QGIS (spatial data/GIS), MATLAB, Python (data processing & analysis), Arduino (C++), LabVIEW, MS Office (financial modeling, NPV/payback analysis), Adobe Creative Cloud
Sustainability & Environmental Tools: EPA National Stormwater Calculator, NREL PVWatts, Low-Impact Development (LID) modeling, Life Cycle Assessment (LCA), carbon emissions analysis
Hardware & Embedded Systems: Analog circuit design, sensor interfacing, op-amps, 555 timers, signal filtering, oscilloscope, Teensy microcontrollers, data logging
Machining & Fabrication: Manual lathe, vertical mill, bandsaw, laser cutter, 3D printing, soldering, heat treatment, precision machining (±0.005 in tolerance)
Engineering Analysis & Design: GD&T, dimensional control, engineering drawing interpretation, iterative prototyping, multi-hazard resilience frameworks

Engineering Projects

Autonomous Surface Vehicle, *Experimental Engineering, Harvey Mudd College, January 2026 - May 2026*

- Led a 4-person engineering team to design, construct, and test an autonomous surface vehicle (ASV) to measure temperature, salinity and depth at Dana Point Harbor.
- Integrated analog sensor-interface circuits utilizing operational amplifiers, 555 timers, and passive filtering to capture analog signals from a thermistor, custom AC-driven conductivity probe, and sonar speaker and microphone.
- Programmed the ASV's controls and datalogging in Arduino C++, and developed MATLAB programs to process raw data into calibrated physical metrics.

Sustainable Wellness Resort and Eco Spa, *Sustainability and Resiliency in the Built Environment, Harvey Mudd College, January 2026 - May 2026*

- Collaborated in a 3-person consulting team to engineer environmental adaptations and financial strategies for a 40-acre upscale coastal resort in a high-risk flood zone.
- Evaluated site hydrology using the EPA's National Stormwater Calculator to model low-impact development (LID) features
- Built Excel models to compare transportation and energy strategies to determine net present value (NPV) and payback periods for proposed scenarios.

Compost Irrigation System, *Intro to Engineering Design and Manufacturing, Harvey Mudd College*
September 2025 - December 2025

- Collaborated in a 5-person engineering team to design, prototype, and test an irrigation system for Sustainable Claremont's compost piles, reducing watering time by ~2 hours per use and improving water efficiency.
- Developed SolidWorks CAD models, created a PVC structural frame, and designed a 3D-printed fluid distribution mechanism.
- Led client meetings, hosted design reviews, and utilized iterative prototyping/

Remotely Operated Vehicle, *Systems Engineering, Harvey Mudd College, September 2025 - December 2025*

- Worked with partner to design and field-test an underwater Remotely Operated Vehicle (ROV) to measure temperature and depth data at the Robert J. Bernard Biological Field Station.
- Developed sensor-interface circuits by soldering a pressure sensor and an thermistor to the project PCB
- Programmed a proportional (P) control loop for motor operation and simulated vertical vehicle dynamics in LabVIEW by dynamically adjusting motor PWM duty cycles to achieve stable depth control.

Hammer Manufacturing Project, *Intro to Engineering Design and Manufacturing, Harvey Mudd College*
September 2025 – November 2025

- Manufactured a precision steel hammer up to ±0.005in tolerance using a manual lathe, vertical mill, bandsaw, sander, and heat treatment processes.
- Interpreted engineering drawings, applied GD&T and dimensional control, and executed the end-to-end machining workflow from raw stock

Home PV System Project, *Personal Project, Kailua, HI*
June, 2025 – July 2025

- Used QGIS to extract rooftop geometry and spatial data, developed AutoCAD structural models, and evaluated solar PV system performance using NREL PVWatts.
- Performed economic and environmental analysis, estimating a 6-year payback period and an annual CO_2 emissions reduction of 2,071 kg, supporting data-driven renewable energy feasibility assessment

Professional Experience

Shift Leader, Teddy's Bigger Burgers, Kailua, HI July, 2022 – August 2025

- Supervised 5 employees per shift, overseeing training of new hires, task delegation, and workflow coordination. Managed order intake and assembly, cash register operations, and end-of-shift cash, while ensuring store closing procedures, cleanliness standards, and conflict resolution

Research Experience

Summer Research, ʻIolani School, Honolulu, HI June, 2022 – August, 2022

- Collaborated with a partner to collect and analyze water samples from 8 surf locations, identifying waterborne bacteria using PCR amplification and gel electrophoresis.

Teaching & Mentorship Experience

Teacher's Assistant, Physics, Punahou School, Honolulu, HI August, 2023 – December, 2023

- Mentored and supported a cohort of 20 students, strengthening problem-solving and analytical skills through guided instruction and feedback. Assisted lab setup, operation, and cleanup, ensuring a safe, organized, and efficient learning environment while supporting hands-on experimentation

Volunteer Camp Counselor, Punahou School, Honolulu, HI October, 2021 – October, 2022

- Led and facilitated group activities for 48 sixth-grade campers during a 3-day overnight camp, maintaining supervision, safety, and engagement. Directly responsible for a cohort of 8 campers, teaching outdoor survival skills, enforcing risk management protocols, and supporting a safe, structured camp environment